1.

Let $B^+ = B(0,1) \cap \{x : x_n > 0\}$, suppose u is harmonic on B^+ and $u = 0$ on $\{x \in \partial B^+ : x_n = 0\}$. Let

$$v(x) = \begin{cases} u(x), & x_n \geqslant 0 \\ -u(x_1, \cdots, -x_n), & x_n < 0 \end{cases}$$

Prove that v is harmonic on $B(0,1)$.
Proof: Note that for any $x \in \partial B^+$ with $x_n = 0$, $\int_{\partial B(x,r)} v(y)\,\mathrm{d}s(y) = 0$ for all small r, so v is continuous and satisfy mean value property. Hence v is harmonic on $B(0,1)$.

2.

Suppose $\Omega \subset \mathbb{R}^n$ open. If $0 \notin \Omega$, denote $x^* = \frac{x}{|x|^2}, \Omega^* = \{x^* : x \in \Omega\}$.
Suppose $u : \Omega \to \mathbb{R}$ and define $K[u] : \Omega^* \to \mathbb{R}, x \mapsto |x|^{2-n} u(x^*)$. Prove that u is harmonic on Ω iff $K[u]$ is harmonic on Ω^*.
Proof: Actually, $\Delta K[u](z) = |z|^{-n-2}(\Delta u)(z^*)$ and $K[K[u]] = u$, so u is harmonic iff $K[u]$ is harmonic.

3.

Suppose u is harmonic on $B = B(0,1)$ and bounded, prove that

$$\sup_{x \in B} (1 - |x|)|Du(x)| < \infty.$$

Proof: Note that Du is also harmonic, so by mean value theorem,

$$Du(x) = \frac{1}{|B(x,r)|} \int_{B(x,r)} Du(y)\,\mathrm{d}y = \frac{1}{|B(x,r)|} \int_{\partial B(x,r)} u(y) \cdot \vec{n}\,\mathrm{d}s(y)$$

which is bounded by $C|\partial B(x,r)|/|B(x,r)| = C'/r$. Let $r = \frac{1-|x|}{2}$, then $B(x,r) \subset B$ so $\sup_{x \in B} (1 - |x|)|Du(x)| < \infty$.

4.

Suppose u is harmonic on $B(0, R_0)$, for $R \in (0, R_0]$ denote $\omega(R) = \sup_{B(0,R)} u - \inf_{B(0,R)} u$.
(1) Prove using Harnack's inequality: there exists $\eta \in (0,1)$ such that $\omega\left(\frac{R}{2}\right) \leqslant \eta\omega(R)$;
(2) If $\sup_{B(0,R_0)} |u| \leqslant M_0$, then there is a constant $\alpha \in (0,1)$ and $C > 0$, such that

$$\omega(R) \leqslant C(M_0 + 1)(R/R_0)^\alpha, R \in (0, R_0].$$

Proof: (1) Let $m_R = \inf_{B(0,R)} u$ and $M_R = \sup_{B(0,R)} u$, then $v(x) = u(x) - m_R$ is harmonic, so by Harnack's inequality $\frac{1}{C} \sup_{B(0,R/2)} v \leqslant \inf_{B(0,R/2)} v$ for some constant C. Hence $\frac{1}{C} M_{R/2} - \frac{1+C}{C} m_{R/2} \leqslant -m_R$, so $\frac{1+C}{C}\omega\left(\frac{R}{2}\right) \leqslant M_{R/2} - m_R \leqslant \omega(R)$.
(2) Note that the constant C is independent of R, since we can scale u to $u(Rx)$, hence η is also independent

of R.
For any $R \in (0, R_0]$, suppose $\frac{R}{R_0} \in (2^{-k}, 2^{-k+1}]$, then $\omega(R) \leqslant \eta\omega(2R) \leqslant \cdots \leqslant \eta^{k-1}\omega(2^{k-1}R)$ with $\frac{2^{k-1}R}{R_0} > \frac{1}{2}$. Hence $\omega(R) \leqslant C(M_0+1)(R/R_0)^\alpha$.

5.

Suppose $\Omega \subset \mathbb{R}^n$ open, $u \in C(\Omega)$. If for any $x \in \Omega$ and $r > 0$ such that $B(x,r) \subset \Omega$, we have

$$u(x) = \frac{1}{|B(x,r)|}\int_{B(x,r)} u(y)\,\mathrm{d}y,$$

then u is harmonic on Ω.
Proof: First suppose $u \in C^2(\Omega)$. Let $g(r) = \frac{1}{|B(x,r)|}\int_{B(x,r)} u(y)\,\mathrm{d}y$, then $g(r) = \frac{1}{\alpha(n)}\int_{B(0,1)} u(x+rv)\,\mathrm{d}v$, so

$$0 = g'(r) = \frac{1}{\alpha(n)}\int_{B(0,1)} v\cdot\nabla u(x+rv)\,\mathrm{d}v = \frac{1}{\alpha(n)}\int_0^1 rt^{n+1}\,\mathrm{d}t\int_{B(0,1)}\Delta u(x+rtz)\,\mathrm{d}z.$$

Therefore if $\Delta u > 0$ in some small neighborhood of x, then $g'(r) > 0$ a contradiction, so u satisfy the mean value property.
Now for general u, take mollifier $f_\varepsilon : \mathbb{R}^n \to \mathbb{R}$, then by dominated convergence, we know that $(u * f_\varepsilon)(x) = \frac{1}{|B(x,r)|}\int_{B(x,r)}(u*f_\varepsilon)(y)\,\mathrm{d}y$. Hence $u * f_\varepsilon$ are all smooth and harmonic, so they satisfy mean value property and so does their uniform limit u. Therefore u is smooth and harmonic.

6.

Suppose $\Omega \subset \mathbb{R}^n$ open, u satisfy

$$\int_\Omega u\Delta\varphi\,\mathrm{d}x = 0, \quad \forall\varphi \in C_c^\infty(\Omega).$$

(a) If $u \in C^2(\Omega)$, prove that u is harmonic on Ω.
(b) If $u \in C(\Omega)$, prove that u is harmonic on Ω.
Proof: (a) If $u \in C^2(\Omega)$, then by Stokes formula,

$$0 = \int_\Omega u\Delta\varphi\,\mathrm{d}x = \int_K u\Delta\varphi\,\mathrm{d}x = \int_K \varphi\Delta u\,\mathrm{d}x$$

so $\int_\Omega \varphi\Delta u\,\mathrm{d}x = 0 \forall\varphi \in C_c^\infty(\Omega)$. Therefore $\Delta u = 0$ and u is harmonic.
(b) If $u \in C(\Omega)$, take a mollifier $f_\varepsilon : \mathbb{R}^n \to \mathbb{R}$, then

$$\begin{aligned}\int_\Omega (u*f_\varepsilon)\Delta\varphi\,\mathrm{d}x &= \int_\Omega\int_{\mathbb{R}^n} u(y)f_\varepsilon(x-y)\Delta\varphi(x)\,\mathrm{d}x\,\mathrm{d}y = \int_{\mathbb{R}^n} u(y)\,\mathrm{d}y\int_\Omega \Delta f_\varepsilon(x-y)\varphi(x)\,\mathrm{d}x \\ &= \int_{\mathbb{R}^n} u(y)(\Delta f_\varepsilon * \varphi)(y)\,\mathrm{d}y = 0\end{aligned}$$

for any $\varphi \in C_c^\infty(\Omega)$ with $\mathrm{supp}\varphi \subset \Omega_\varepsilon$ where $\Omega_\varepsilon = \{x \in \Omega : d(x, \Omega^C) > \varepsilon\}$, since $\Delta f_\varepsilon * \varphi \in C_c^\infty(\Omega)$. Therefore each $u * f_\varepsilon$ is harmonic on Ω_ε, hence u is harmonic.